

A Doubling Non-Differentiability Space with Rank-One Cotangent Module

Automated Math Discovery Candidate Proof

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Status. Candidate full affirmative solution, likely valid pending expert review. For every $1 < p < \infty$, the example below is compact and doubling, is not a Lipschitz differentiability space, and has nonzero L^p -cotangent module of constant local dimension one.

Source and Open Question

This packet addresses the unnumbered question on page 25 of Martin Kell's paper *On Cheeger and Sobolev differentials in metric measure spaces* [1], arXiv:1512.00828. The question is:

Is there a metric measure space $(M, d, \mathbf{m})$ with $\mathbf{m}$ being (locally) doubling that is not a differentiability space but whose L^p -cotangent module is non-trivial and locally finite dimensional bounded by some $N \in \mathbb{N}$?

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trivial construction is as follows: Let M be a differentiability space with trivial L^p -cotangent module, i.e. $L^p_{\text{lip}_0}(T^*M) = L^p_{\text{lip}}(T^*M)$: Let $([0, 1], 2^{-i}d_{\text{Euclid}}, \mathbf{m}_i)$ where $\mathbf{m}_i = 2^{-i}w\lambda$ where w is the DiMarino-Speight $A_{p'}$ -weights on $[0, 1]$ with $p < p'$ (see [DS15]). Then the L^p -product of those spaces is not a differentiability spaces in the sense above but its L^p -structure is trivial. However, the product measure is not doubling. As metrically doubling have reflexive Sobolev spaces [ACD12] we can ask the following.

Question. *Is there a metric measure space $(M, d, \mathbf{m})$ with $\mathbf{m}$ being (locally) doubling that is not a differentiability space but whose L^p -cotangent module is non-trivial and locally finite dimensional bounded by some $N \in \mathbb{N}$?*

The following is the central result of Cheeger's paper on a generalized Rademacher Theorem ([Che99], see also [FHK99, Theorem 10]). We state it in the context of the theory developed in this paper. We refer the reader to the corresponding papers for exact definitions. Compare the result also to [Gig14, Corollary 2.5.2].

Theorem 2.31. *Assume $(M, d, \mathbf{m})$ is a metric measure space satisfying locally and*

Definitions and Construction

Let $\mathcal{C} \subset [0, 1]$ be the middle-thirds Cantor set and let ν be the standard Cantor probability measure. Put

$$M = [0, 1] \times \mathcal{C}, \quad d((x, y), (x', y')) = \sqrt{|x - x'|^2 + |y - y'|^2}, \quad \mathbf{m} = \mathcal{L}^1 \llcorner [0, 1] \otimes \nu.$$

Fix $1 < p < \infty$. For $f \in \text{Lip}(M)$ write

$$\text{lip } f(z) = \limsup_{w \rightarrow z} \frac{|f(w) - f(z)|}{d(w, z)}.$$

The relaxed p -energy used to construct the L^p -cotangent module is

$$\mathcal{E}_p(f) = \inf \left\{ \liminf_{k \rightarrow \infty} \int_M (\text{lip } f_k)^p d\mathbf{m} : f_k \in \text{Lip}(M), f_k \rightarrow f \text{ in } L^p(\mathbf{m}) \right\}. \quad (1)$$

Its minimal relaxed gradient is denoted $|D_p f|$, so that $\mathcal{E}_p(f) = \int |D_p f|^p d\mathbf{m}$ on its domain.

Proof Intuition

The Cantor factor is visible to the metric but invisible to the relaxed Sobolev energy. At every finite Cantor scale, a function which is smooth in x and constant on each clopen Cantor cylinder is globally Lipschitz. Its pointwise slope is exactly the horizontal derivative. Such functions are dense in $L^p(\mathcal{C}; W^{1,p}([0, 1]))$. Conversely, every Lipschitz approximation controls its horizontal derivative by its pointwise slope. Consequently

$$|D_p f| = |\partial_x f|, \quad D(\mathcal{E}_p) = L^p(\mathcal{C}, \nu; W^{1,p}([0, 1])).$$

Thus dx generates the whole cotangent module and the module has rank one.

The same scale separation prevents a differentiability structure. The Cantor-coordinate function $h(x, y) = y$ has vertical difference quotients equal to one at every point. On the other hand, its derivative along every Lipschitz curve fragment is zero, because the Cantor set has Lebesgue measure zero. The necessary direction of Bate's characterization supplies a universal independent collection of Alberti representations on countably many pieces [2, Theorem 7.8]. The zero vector is then a gradient of h along each collection, so the theorem identifies its chart differential with zero, contradicting the vertical difference quotients.

Main Result

Theorem 1. *For every $1 < p < \infty$, the compact metric measure space $(M, d, \mathbf{m})$ above has the following properties.*

1. *The measure $\mathbf{m}$ is doubling (indeed, it is Ahlfors $1 + \log 2 / \log 3$ regular).*
2. *Its relaxed energy and domain are*

$$D(\mathcal{E}_p) = L^p(\mathcal{C}, \nu; W^{1,p}([0, 1])), \quad \mathcal{E}_p(f) = \int_M |\partial_x f|^p d\mathbf{m}. \quad (2)$$

Equivalently, $|D_p f| = |\partial_x f|$ almost everywhere.

3. *Its L^p -cotangent module is isometrically $L^p(M, \mathbf{m}) dx$. In particular it is nontrivial and has constant local dimension one.*
4. *The space is not a Lipschitz differentiability space.*

Therefore $(M, d, \mathbf{m})$ answers Kell's question affirmatively, with the uniform bound $N = 1$.

The Energy Computation

Put $s = \log 2 / \log 3$. The standard Cantor measure is Ahlfors s -regular: there are constants $0 < c < C < \infty$ such that

$$cr^s \leq \nu(B_{\mathbb{R}}(y, r) \cap \mathcal{C}) \leq Cr^s \quad (y \in \mathcal{C}, 0 < r \leq 1). \quad (3)$$

This follows directly by choosing n with $3^{-(n+1)} < r \leq 3^{-n}$ and comparing the ball with the level- n and level- $(n+1)$ Cantor cylinders. Euclidean balls in the product are sandwiched between max-metric product balls with radii $r/\sqrt{2}$ and r . Together with $\mathcal{L}^1([x-r, x+r] \cap [0, 1]) \asymp r$, this proves that $\mathbf{m}$ is Ahlfors $1+s$ -regular and hence doubling.

Lemma 1. *The identity (2) holds.*

Proof. First prove the lower bound. Let $f_k \in \text{Lip}(M)$ converge to f in $L^p(\mathbf{m})$ and suppose

$$\sup_k \int_M (\text{lip } f_k)^p d\mathbf{m} < \infty.$$

For every k , Fubini's theorem and one-dimensional Rademacher differentiability on the horizontal segments give

$$|\partial_x f_k(x, y)| \leq \text{lip } f_k(x, y) \quad \text{for } \mathbf{m}\text{-almost every } (x, y). \quad (4)$$

Since $1 < p < \infty$, after passing to a subsequence we may assume that $\partial_x f_k$ converges weakly in $L^p(\mathbf{m})$ to some g . Testing against smooth functions in x and bounded measurable functions in y shows

$$\int_M f \partial_x \psi d\mathbf{m} = - \int_M g \psi d\mathbf{m}.$$

Thus $f \in L^p(\mathcal{C}; W^{1,p}([0, 1]))$ and $g = \partial_x f$. Weak lower semicontinuity and (4) yield

$$\int_M |\partial_x f|^p d\mathbf{m} \leq \liminf_{k \rightarrow \infty} \int_M (\text{lip } f_k)^p d\mathbf{m}. \quad (5)$$

In particular, finite relaxed energy forces membership in the displayed Bochner–Sobolev space.

For the converse, let $\mathcal{P}_n$ be the partition of $\mathcal{C}$ into its 2^n clopen level- n cylinders. Conditional expectation in the Cantor variable gives

$$F_n(x, y) = \sum_{Q \in \mathcal{P}_n} u_Q(x) \mathbf{1}_Q(y), \quad u_Q(x) = \frac{1}{\nu(Q)} \int_Q f(x, z) d\nu(z).$$

The cylinder sigma-algebras increase to the Borel sigma-algebra of $\mathcal{C}$. The vector-valued martingale convergence theorem, applied in the separable space $W^{1,p}([0, 1])$, gives

$$F_n \longrightarrow f \quad \text{in } L^p(\mathcal{C}; W^{1,p}([0, 1])). \quad (6)$$

For fixed n , approximate the finitely many u_Q in $W^{1,p}([0, 1])$ by smooth functions. A diagonal sequence therefore reduces matters to functions of the form

$$F(x, y) = \sum_{Q \in \mathcal{P}_n} v_Q(x) \mathbf{1}_Q(y), \quad v_Q \in C^\infty([0, 1]). \quad (7)$$

Every such F is globally Lipschitz on M : the cylinders are finitely many pairwise disjoint clopen sets with positive mutual separation. Locally in the Cantor variable F is constant, and hence, at every point where v_Q is differentiable,

$$\text{lip } F(x, y) = |v'_Q(x)| = |\partial_x F(x, y)|. \quad (8)$$

The convergence in (6), the smooth approximation, and (8) show that every $f \in L^p(\mathcal{C}; W^{1,p}([0, 1]))$ admits Lipschitz approximants F_k with

$$F_k \rightarrow f \text{ in } L^p(\mathfrak{m}), \quad \partial_x F_k \rightarrow \partial_x f \text{ in } L^p(\mathfrak{m}), \quad \int (\text{lip } F_k)^p d\mathfrak{m} \rightarrow \int |\partial_x f|^p d\mathfrak{m}.$$

This is the reverse inequality to (5), and proves both the domain and energy identities. Equality of the minimal relaxed gradient with $|\partial_x f|$ follows from the local version of the same lower and upper bounds (or by multiplying the construction by cylinder-simple cutoff functions). $\square$

Identification of the Cotangent Module

The L^p -cotangent module is obtained by completing piecewise differentials with pointwise norm $|D_p f|$ as in Kell's construction [1]. By Lemma 1, the assignment

$$d_p f \mapsto \partial_x f \quad (9)$$

preserves pointwise norms and is compatible with the defining equivalence relation. It therefore extends to an isometric module map into $L^p(M, \mathfrak{m})$.

For the coordinate function $X(x, y) = x$, Lemma 1 gives $|d_p X| = 1$ almost everywhere. Write $dx = d_p X$. Multiplication of dx by bounded measurable functions, followed by L^p completion, produces all of $L^p(M, \mathfrak{m}) dx$. Hence (9) is onto and

$$L^p(T^*M) \cong L^p(M, \mathfrak{m}) dx.$$

The module is consequently nonzero and has local dimension exactly one at $\mathfrak{m}$ -almost every point.

Failure of Lipschitz Differentiability

Let

$$h : M \rightarrow \mathbb{R}, \quad h(x, y) = y.$$

This function is 1-Lipschitz. We first record what it does on curve fragments. If $K \subset \mathbb{R}$ is compact and $\gamma = (\gamma_1, \gamma_2) : K \rightarrow M$ is Lipschitz, the one-dimensional area formula gives

$$\int_K |\gamma'_2(t)| dt = \int_{\mathbb{R}} N(\gamma_2, K, z) dz = 0,$$

because $\gamma_2(K) \subset \mathcal{C}$ and $\mathcal{L}^1(\mathcal{C}) = 0$. Therefore

$$(h \circ \gamma)' = \gamma'_2 = 0 \quad \text{almost everywhere on } K. \quad (10)$$

Suppose, toward a contradiction, that $(M, d, \mathfrak{m})$ were a Lipschitz differentiability space. The necessary direction of Bate's characterization [2, Theorem 7.8] gives a countable Borel decomposition, up to a null set, into pieces U carrying finite universal collections of φ -independent Alberti

representations. Each (U, φ) is a chart, and the chart differential of a Lipschitz function is given by any gradient along the collection. At almost every relevant curve point, such a gradient G_h satisfies

$$(h \circ \gamma)'(t) = G_h(\gamma(t)) \cdot (\varphi \circ \gamma)'(t). \quad (11)$$

The left side is zero by (10), so $G_h = 0$ is a gradient of h along every collection. The last assertion of Bate’s theorem therefore gives $D_\varphi h = 0$ almost everywhere on every positive-dimensional piece. If zero-dimensional charts are allowed, their differential is zero by definition and the same conclusion holds.

However, $\mathcal{C}$ is perfect. At every $(x, y) \in M$ one can choose distinct $y_j \in \mathcal{C}$ with $y_j \rightarrow y$, and then

$$\frac{|h(x, y_j) - h(x, y)|}{d((x, y_j), (x, y))} = 1. \quad (12)$$

Thus h cannot satisfy the chart differentiability condition with zero differential at any point. This contradicts the preceding conclusion on every positive-measure chart. Hence no countable differentiability atlas can cover M up to a null set, and $(M, d, \mathfrak{m})$ is not a Lipschitz differentiability space. The proof of Theorem 1 is complete.

Verification Notes and Limitations

The proof is analytic and uses no computer-assisted mathematical step. The following checks were made during preparation.

- The relaxation was proved in both directions, including identification of its full domain; it is not only a calculation on Lipschitz functions.
- The non-differentiability argument is formulated for Lipschitz curve fragments. The area formula makes their Cantor-coordinate derivative vanish.
- The conclusion is stronger than requested: one compact globally doubling example works for every $p \in (1, \infty)$, and the dimension bound is $N = 1$.
- The endpoint $p = 1$ is not claimed. Reflexive weak compactness in the lower energy estimate was used for $1 < p < \infty$, which is also the range in the surrounding discussion of the source question.

Bounded Novelty Check

On July 22, 2026, the run’s registry, solution, attempt, proof-gap, title, and source-ranked indexes were searched for arXiv:1512.00828 and combinations of “Cantor,” “cotangent module,” “Sobolev module,” and “differentiability space.” They contained no prior answer. Bounded web searches used the exact source title and the phrases “middle-thirds Cantor cotangent module,” “Cantor measure cotangent module Sobolev,” “not a differentiability space rank one cotangent module,” and “On Cheeger and Sobolev differentials Cantor product.” The searches found the source, publication and citation pages, and unrelated module papers, but no direct answer or occurrence of this product example. Novelty confidence is moderate pending a specialist literature review.

Human Review Recommendation

This packet is recommended for expert review as a candidate full solution. The main points to check are: (i) that the source paper's relaxed cotangent module convention matches (1) without an inessential normalization; (ii) the vector-valued cylinder martingale approximation and localized minimal-gradient identity; and (iii) the precise use of Bate's Theorem 7.8 and the chart chain rule in deriving $D_\varphi h = 0$.

References

- [1] M. Kell, *On Cheeger and Sobolev differentials in metric measure spaces*, arXiv:1512.00828; Rev. Mat. Iberoam. **35** (2019), no. 7, 2119–2150.
- [2] D. Bate, *Structure of measures in Lipschitz differentiability spaces*, arXiv:1208.1954; J. Amer. Math. Soc. **28** (2015), no. 2, 421–482.